Python Strings

Python Multiline Strings

```
# Multiline string
message = """To avoid pain, they avoid pleasure.
To avoid death, they avoid life."""

print(message)
```

```
To avoid pain, they avoid pleasure.
To avoid death, they avoid life.
```

Access String Characters

```
model = 'ChatGPT'

# Access the first character
print(model[0])    # Output: C

# Access the fifth character
print(model[4])    # Output: G
```

```
C
G
```

negative indexing

```
model = 'ChatGPT'

# Access the last character
print(model[-1])   # Output: T

# Access the fourth last character
print(model[-4])   # Output: t
```

```
T
t
```

slicing

```
model = 'ChatGPT'

# Access characters from index 0 up to (but not including) 4
print(model[0:4])   # Output: Chat
```

```
Chat
```

Strings are Immutable

```
# model = 'ChatGPT'

# model[0] = 'W'   TypeError: 'str' object does not support item assignment
# print(model)

model = 'Opus'
version = '5'

model = model + " " + version
print(model)    # Opus 5
```

```
Opus 5
```

Python String Methods

```
text = "ChatGPT is great."

# Replace "ChatGPT" with "Claude"
new_text = text.replace("ChatGPT", "Claude")

print(new_text)  # Output: Claude is great.
```

```
Claude is great.
```

String Membership Test

```
print('Chat' in 'ChatGPT')        # True
print('Claude' not in 'ChatGPT')  # True
```

```
True
True
```

Iterate Through a String

```
model = 'Opus'

for c in model:
    print(c)
```

```
O
p
u
s
```

Python String Length

```
model = 'Opus'

# Count the number of characters
print(len(model))   # Output: 4
```

```
4
```

Escape Sequences

```
# example = "He said, "What's there?""

# print(example) # Error


# escape double quotes
example = "He said, \"What's there?\""
```

```
# escape single quotes
example = 'He said, "What\'s there?"'

print(example)

# Output: He said, "What's there?"
```

```
He said, "What's there?"
```

String Formatting (f-Strings)

```
company = 'Google'
field = 'AI'

message = f'{company} is an {field} company.'
print(message)
```

```
Google is an AI company.
```